

## 6. Bootstrap Program aur Booting Process

Computer Science Fundamentals | Module 2: System Booting | CS\_Fundamental-00

**Reference to Previous Concepts:** Concept 4.1 mein humne dekha tha ki Kernel OS ka core engine hai jo boot par RAM mein load hota hai. Yeh topic **Image 6 & Image 7** par aadharit hai jo batata hai ki RAM khaali hone par Kernel ko kaun aur kaise load karta hai!

### 6.1 Bootstrap Program (Bootstrap Loader) kya hai?

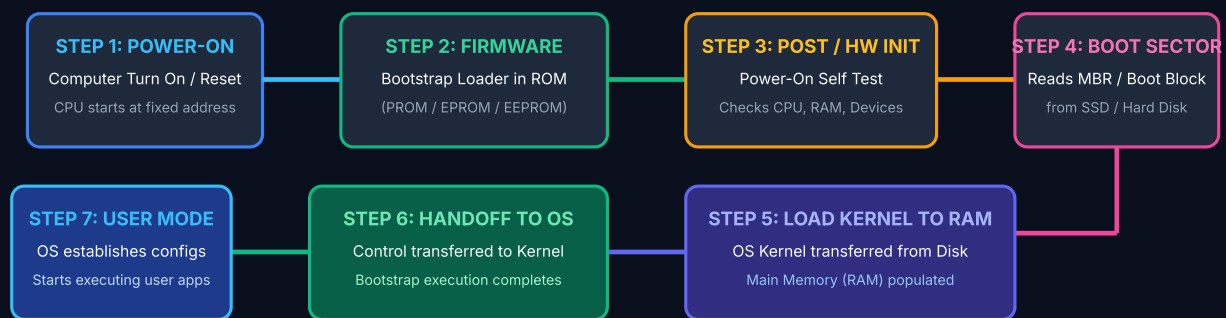
- **Initial Startup Code:** Chhota sa code jo non-volatile **Firmware / ROM** mein store hota hai.
- **Primary Purpose:** System hardware initialize karna aur storage drive se OS Kernel ko Main Memory (RAM) mein load karke CPU ka control usse sonpna.

### 6.2 Firmware & ROM Types (Image 6 Handwritten Note)

- **ROM:** Permanent Read-Only Memory.
- **PROM / EPROM:** Programmable aur UV-erasable ROM chips.
- **EEPROM / Flash:** Electrically Erasable Programmable ROM (Modern BIOS / UEFI chips).

#### Fig 6.1: Bootstrap Process & System Booting Sequence

Reference: Image 6 & 7 [Firmware, POST, Boot Block, Kernel Handoff]



#### Summary of Firmware Terminology (Image 6 Note):

**ROM (Read Only Memory):** Permanent non-volatile storage jisme initial bootstrap code rehta hai. **PROM / EPROM / EEPROM:** Programmable / Erasable ROM chips jisme firmware updates flash kiye jate hain (BIOS / UEFI). **Boot Sector / Boot Block:** Storage device ka first physical sector (MBR/GPT) jo OS bootloader code rakhta hai. **Kernel Handoff:** Ek baar Kernel RAM mein load hua, firmware CPU ka control Kernel ko de deta hai aur kaam khatam kar deta hai.

## 6.3 Step-by-Step Working of Bootstrap Program (Images 6 & 7)

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1. **Power-On or Reset:** CPU fixed startup routine se instruction fetch karna shuru karta hai.
2. **POST (Hardware Diagnostics):** CPU registers, memory aur peripheral devices ki integrity check hoti hai.
3. **Loading the OS:** Boot device (SSD/HDD) ke **Boot Sector / Boot Block** ko read karke Kernel ko RAM mein transfer karta hai.
4. **Handoff to Kernel:** Kernel RAM mein aate hi CPU execution control Kernel ko handoff kar diya jata hai.
5. **User Mode Transition:** OS system configurations establish karta hai aur user applications chalane ke liye **User Mode** mein enter kar jata hai.